Using Machine Learning to Optimize Hybrid Multimodal Travel Routing in Smart Cities

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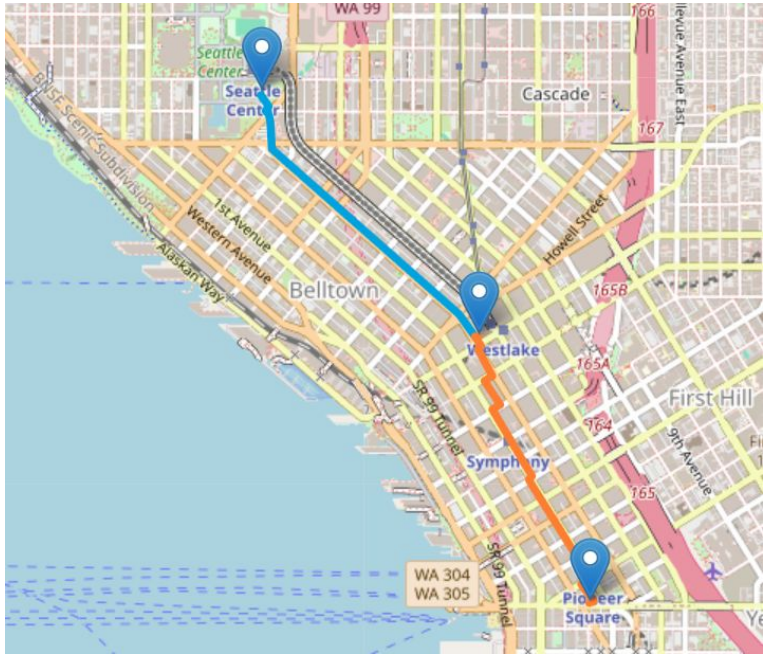
Motivation



Urban issue: Rising congestion + emissions + inefficient single-mode routing.

Observation: Tools like Google Maps optimize only one mode (driving).

Goal: Design a framework that balances speed and sustainability across driving, transit, bicycling, and walking.

Background

- Classical methods: Shortest Path Algos → limited scalability.
- Machine Learning advances: RF for prediction, RL for adaptive planning.
- Gaps: Prior RL approaches often simulation-based, single-objective.

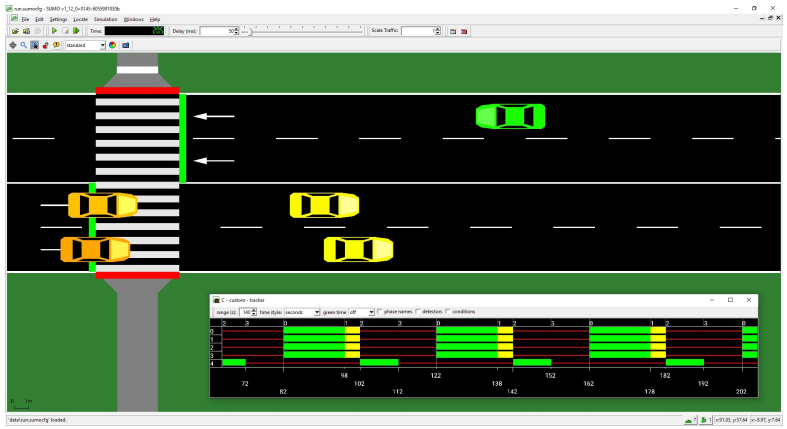
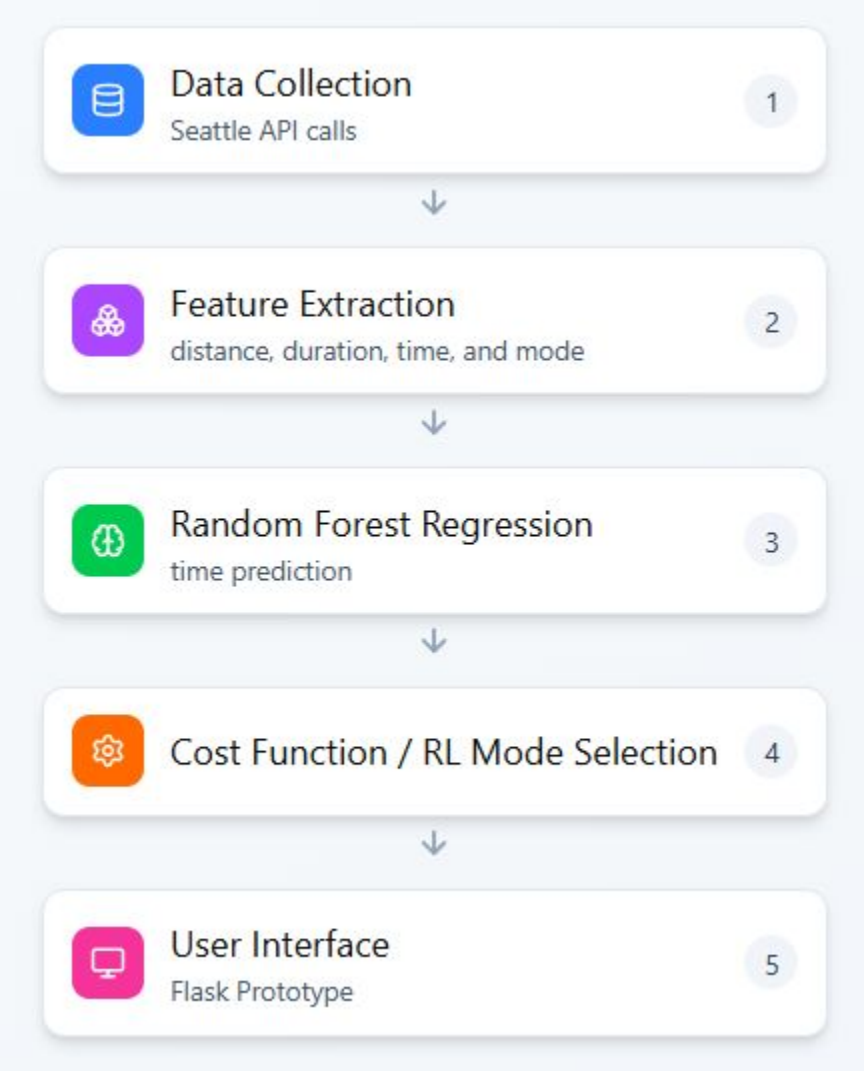


Image of Simulation of Urban MObility

Research Objectives + Pipeline

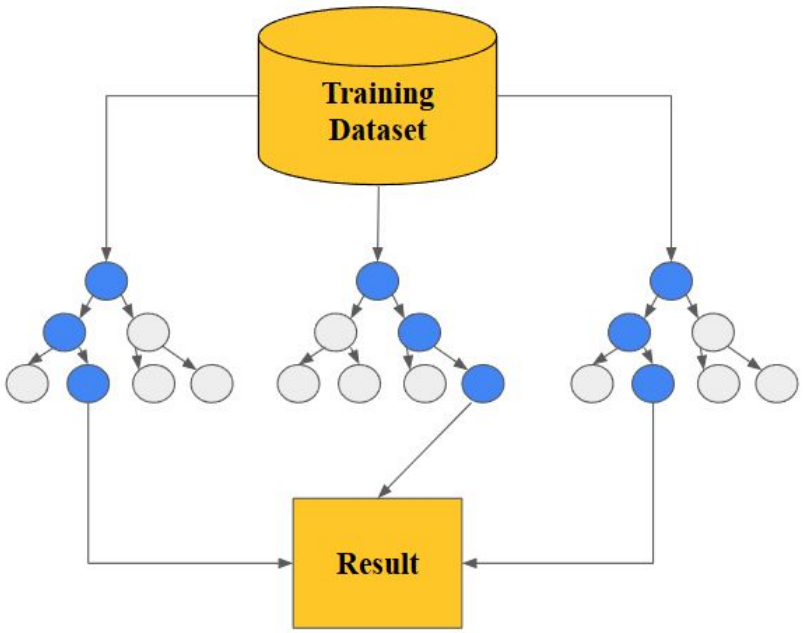


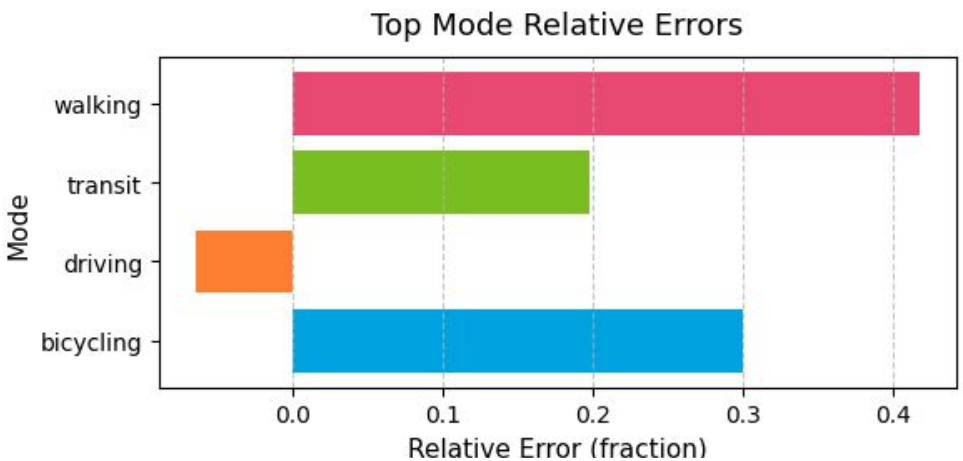
Data Collection

Collected from 6 AM–10 PM for 7 days in Aug 2025, for .
52 origin-destination pairs × 4 modes → 24,752 records.

Timestamp	Origin	Destination	Mode	Distance (m)	Duration (s)	Duration in Traffic (s)
1754312400	Space Needle	Westlake Park	Driving	2010	540	447
1754312400	Space Needle	Westlake Park	Transit	1981	931	—
1754312400	Space Needle	Westlake Park	Walking	1505	1286	—
1754312400	Space Needle	Westlake Park	Bicycling	1757	490	—

Random Forest Regression





- Low mean error for driving (**-6.5%**)
- Moderate mean error for Transit (**19.7%**)
- High mean errors for bicycling (**29.9%**) and walking (**41.8%**)

Q Learning (RF Algorithm)

Mode	Count
Driving	93
Walking	13
Transit	11
Bicycling	2

Q-Learning Routes from a Sample Week

- Agent learned mode choice based on reward = time + emission trade-off.
- Despite parameter tuning, driving dominated (**~78%** of selections).
- Reward sparsity and dataset bias → limited diversity

Cost Function

$$\text{Cost} = u \cdot \frac{t}{T} + v \cdot \frac{e}{E} + \text{Time Penalty} + \text{Emissions Penalty}$$

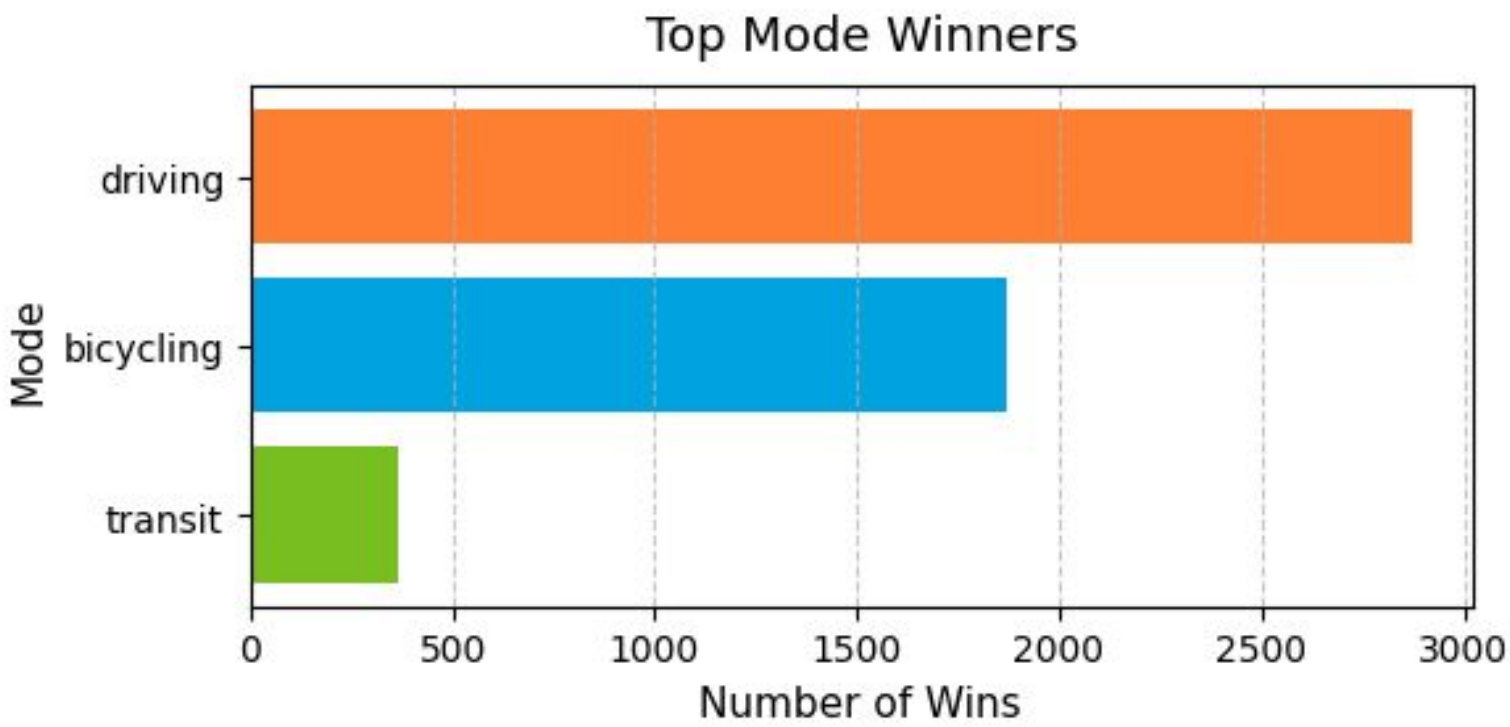
$$\text{Time Penalty} = \alpha \cdot \max(0, t - T)$$

$$\text{Emissions Penalty} = \beta \cdot \max(0, e - E)$$

t : actual travel time of route (seconds)
T : max acceptable travel time (constraint)
e : estimated emissions of route (g CO_2)
E : max acceptable emissions (constraint)
u,v : weighting factors for time and emissions in the baseline cost
α,β : penalty coefficients for exceeding travel time and emissions

$T = 1800$
$E = 1000$
$u = 15$
$v = 2.5$
$\alpha = 0.04$
$\beta = 0.015$

Cost Function Results



- Driving had 2,876 wins (**~56%**)
- Bicycling had 1,873 wins (**~37%**)
- Transit had 368 wins (**~7%**)

User Interface Demo

Travel Time Predictor

Place A (Origin) — Coordinates (lat,lng):
47.620475, -122.349226

Place B — Coordinates (lat,lng):
47.610949, -122.337020

Place C (End) — Coordinates (lat,lng):
47.601848, -122.331854

Mode (used for both legs in "Predict"):
Driving

Hour (6–22): 14
Day of Week: Friday

Predict | Best Modes (Per Leg)

Recommended Modes: A→B: **Bicycling**, B→C: **Driving**

Leg A→B (Bicycling) — distance 1401.35 m

Time: 490.0 s | Emissions: 0.0 g CO_2 | Score: 4.08

Leg B→C (Driving) — distance 1083.87 m

Time: 306.13 s | Emissions: 208.1 g CO_2 | Score: 3.07

Total Predicted Time: 796.13 seconds

Total Estimated Emissions: 208.1 g CO_2

Total Route Score: 7.15

Travel Time Predictor

Place A (Origin) — Coordinates (lat,lng):
47.620475, -122.349226

Place B — Coordinates (lat,lng):
47.610949, -122.337020

Place C (End) — Coordinates (lat,lng):
47.601848, -122.331854

Mode (used for both legs in "Predict"):
Driving

Hour (6–22): 15
Day of Week: Friday

Predict | Best Modes (Per Leg)

Recommended Modes: A→B: **Bicycling**, B→C: **Transit**

Leg A→B (Bicycling) — distance 1401.35 m

Time: 490.0 s | Emissions: 0.0 g CO_2 | Score: 4.08

Leg B→C (Transit) — distance 1083.87 m

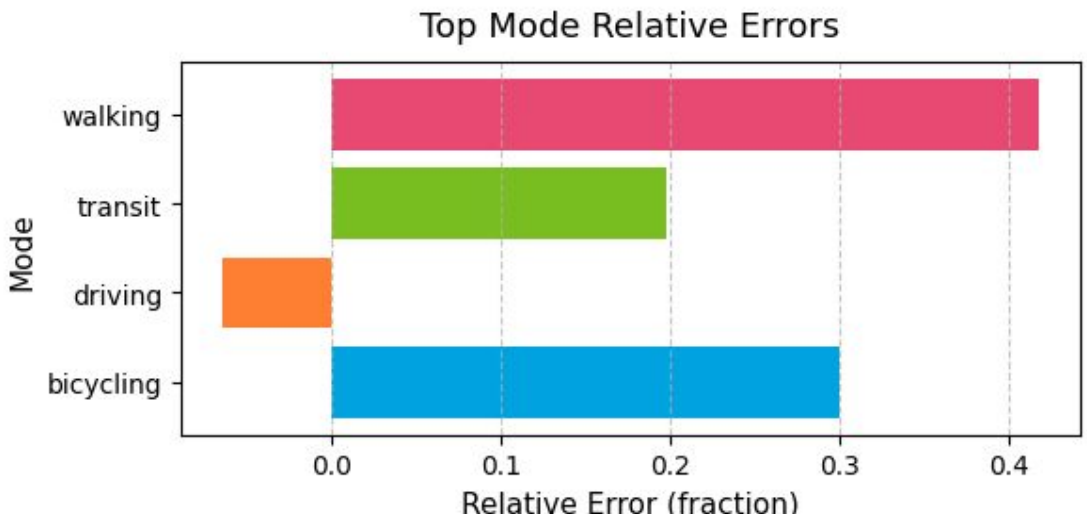
Time: 357.29 s | Emissions: 113.81 g CO_2 | Score: 3.26

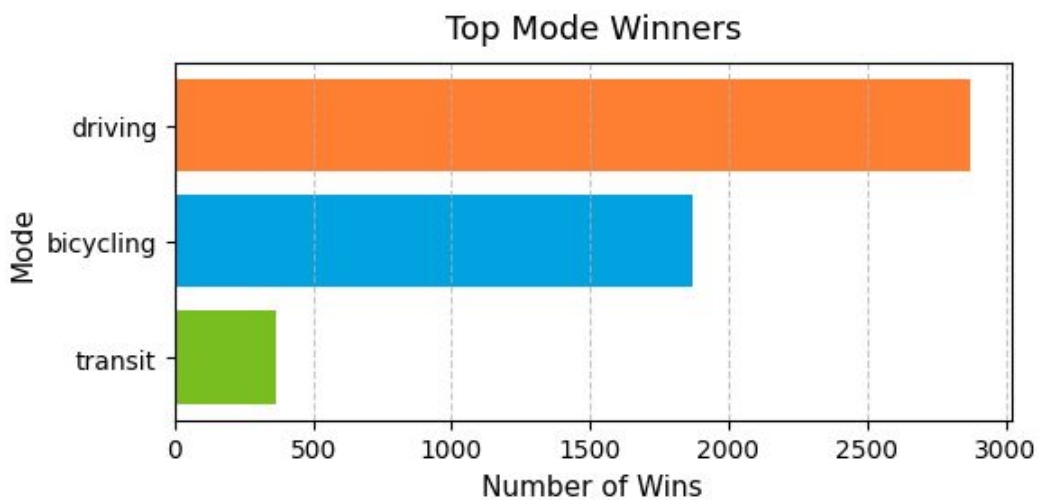
Total Predicted Time: 847.29 seconds

Total Estimated Emissions: 113.81 g CO_2

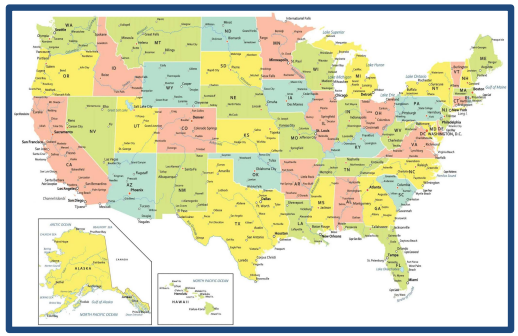
Total Route Score: 7.35

Key Findings





Smart City Impact



- Aligns with UNLV Smart Cities vision.
- Can be extended to other cities with local data.

Future Work

- Expand dataset to include weather and energy cost.
- Improve walking/biking accuracy via additional temporal data.
- Explore Deep Q-Learning and multi-agent RL.
- Deploy as mobile app prototype for real users.

Conclusion

- Hybrid optimization model → fast + sustainable routes
- Weighted cost + penalties balance time & emissions
- Seattle results: mode choice shifts by day / hour / limits
- Supports Smart-City mobility without user inconvenience
- Next steps: richer emissions models, new modes, scaling to more cities

Acknowledgements